

Lab 1:

① adding 1st loop:

```
for (int i=0; i<n; i++) {  
    for (int j=i; j<n; j++) {
```

→ $i=0 \rightarrow n$ times $j=0, 1, 2, 3, 4 \rightarrow 4$ times

$i=1 \rightarrow n-1$ times $j=1, 2, 3 \rightarrow 3$ times

$i=2 \rightarrow n-2$ times $j=2, 3 \rightarrow 2$ times

$i=3 \rightarrow n-3$ times $j=3 \rightarrow 1$ times

$4 + 3 + 2 + 1 = 10$ is $n(n-1) + (n-2) + (n-3) + \dots + 1$

$n(n+1)/2 = 2n^2/2 = 2$ (Drop constant & also drop

Big O = $O(n^2)$ lower order term).

② for (int i = 2; i < n; i = i * i) {
}

if $n = 16$, 2, 4, ~~8~~, 16
 $i = 16 \neq 16$

only 2 iterations & the value is being squared ($i \times i$)

$\therefore O(\log \log n)$

This grows more slowly than $O(\log n)$.

② Lab 4:

1. Duplicate Check.java:

1. Slow version (has Duplicate Nested) is $O(n^2)$ because it compares every pair of elements, while the
2. (HashSet) version is $O(n)$ on average because each element is checked/inserted in $O(1)$ average time.

2. String Concat Trap.java:

1. String += is $O(n^2)$ because immutable string repeatedly copy the existing characters.
2. While the ~~HashSet~~ StringBuilder is $O(n)$ because append is amortized $O(1)$.

3. List Trap.java:

ArrayList.remove() is $O(n^2)$ because every removal shifts the remaining elements, whereas ArrayDeque.poll() is $O(n)$ overall because each removal is $O(1)$.

③ Home Challenge 1:

1. int[] arr = {10, 5, 20, 15, 30};

2. after sorting - $\{5, 10, 15, 20, 30\}$

3. arr.length = 5;

if $k=2$

$\boxed{\text{arr.length} - k}$ - To find the k -largest no.
 $= 5 - 2$
 $= 3$

4. Index: 0 1 2 3 4

values: 5 10 15 20 30

$\therefore$ 3rd index = 20 value

5. So result = 20 is k -th largest.

largest = 30 $\rightarrow k=1$

2nd largest = 20 $\rightarrow k=2$

3rd largest = 15 $\rightarrow k=3$

Complexity : $O(n \log n)$

$O(n \times k)$ is better when k is small.

$O(k \log n)$ is better when k is large

Home challenge 2:

arr = $\{9, 7, 11, 15\}$

target = 18

$9 + 7 = 16$, $16 \neq 18$ so false

$7 + 11 = 18$, $18 = 18$ so true.

Complexity : $O(n^2)$ - nested loop